

Program 6

Docker Image Build, Push to Docker Hub & GitHub Container Registry

1. Create project directory

```
mkdir program-6 && cd program-6
```

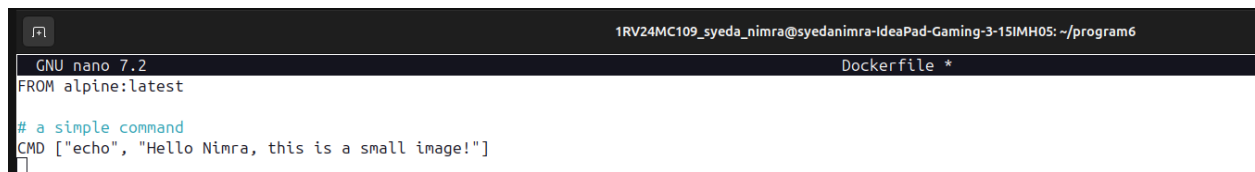
Step 1: Create a Dockerfile

```
nano Dockerfile
```

```
FROM alpine:latest
```

```
# a simple command
```

```
CMD ["echo", "Hello Nimra, this is a small image!"]
```

A screenshot of a terminal window with a dark background. The title bar shows the user '1RV24MC109_syeda_nimra' and the path '~/program6'. The terminal content shows the 'nano' editor editing 'Dockerfile'. The text in the editor is: 'FROM alpine:latest', '# a simple command' (highlighted in cyan), and 'CMD ["echo", "Hello Nimra, this is a small image!"]'.

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05: ~/program6
GNU nano 7.2 Dockerfile *
FROM alpine:latest
# a simple command
CMD ["echo", "Hello Nimra, this is a small image!"]
```

Step 2: Build the Docker Image

```
docker build -t program-6 .
```

```

1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/program6$ docker build -t program6 .
[+] Building 9.7s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 128B
=> [internal] load metadata for docker.io/library/alpine:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/1] FROM docker.io/library/alpine:latest@sha256:865b95f46d98cf867a156fe4a135ad3fe50d2056aa3f25ed31662dff6da4eb62
=> => resolve docker.io/library/alpine:latest@sha256:865b95f46d98cf867a156fe4a135ad3fe50d2056aa3f25ed31662dff6da4eb62
=> => sha256:865b95f46d98cf867a156fe4a135ad3fe50d2056aa3f25ed31662dff6da4eb62 9.22kB / 9.22kB
=> => sha256:1882fa4569e0c591ea092d3766c4893e19b8901a8e649de7067188aba3cc0679 1.02kB / 1.02kB
=> => sha256:e7b39c54cdeca0d2aae83114bb605753a5f5bc511fe8be7590e38f6d9f915dad 611B / 611B
=> => sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 3.86MB / 3.86MB
=> => extracting sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612
=> => exporting to image
=> => exporting layers
=> => writing image sha256:0a2a0572b83655ffb59bfc864894f0efcaaed3b1698c31c62908b613c67c65d9
=> => naming to docker.io/library/program6
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/program6$

```

Step 3: Tag the Image for Docker Hub

docker tag program-6 <dockerhub-username>/program-6:latest

```

=> => naming to docker.io/library/program6
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/program6$
docker tag program6 nimra07/program6:latest
1RV24MC109_syeda_nimra@svedanimra-IdeaPad-Gaming-3-15IMH05:~/program6$ docker

```

Step 4: Push the Image to Docker Hub

Login to Docker Hub using your Docker Hub credentials (Personal Access Token as password).

docker login -u <dockerhub-username>

Push the image to Docker Hub:

docker push <dockerhub-username>/program-6:latest

```
docker tag program6 nimra07/program6:latest
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/program6$ docker login -u nimra07

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password: 
```

Step 5: Tag the Image for GitHub Container Registry (GHCR)

Tag the same image for GitHub Container Registry.

```
docker tag program-6 ghcr.io/<github-username>/program-6:latest
```

Step 6: Push the Image to GHCR

Login to GHCR using GitHub Personal Access Token (PAT).

```
echo $GITHUB_PAT | docker login ghcr.io -u <github-username> --password-stdin
```

Push the image:

```
docker push ghcr.io/<github-username>/program-6:latest
```

Step 7: Connect the Local Project to GitHub (Source Code)

```
git init
git remote add origin https://github.com/<github-username>/program-6.git
git remote set-url origin https://github.com/<github-username>/program-6.git
git remote -v
```

Step 8: Push the Source Code to GitHub

```
git add .
```

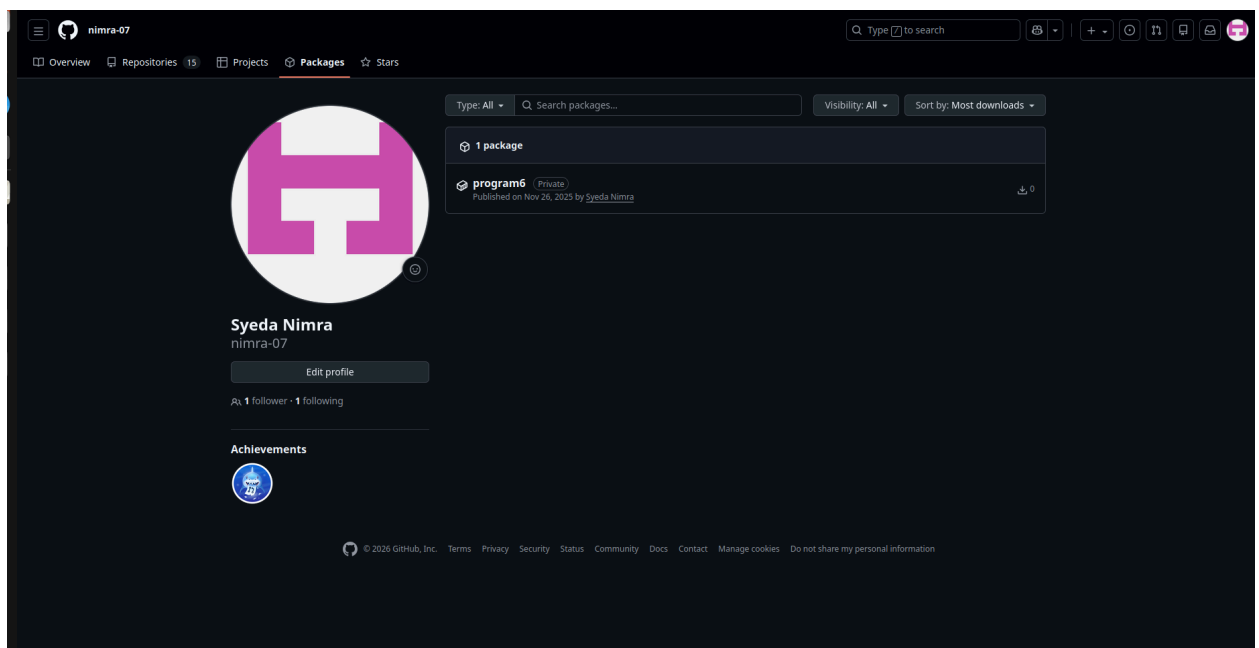
```
git commit -m "Initial commit"
```

```
git push -u origin main
```

Step 9: Verify Uploads on Both Registries

GitHub

- Ensure the repository program-6 exists on GitHub.
- Navigate to Packages.



Docker Hub

- Login to Docker Hub.
- Go to **Repositories**.

New Docker Hardened Images are now FREE for every developer. Read blog. →

hub

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nimra07

Docker Personal

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Usage

Pulls

Storage

Repositories

All repositories within the nimra07 namespace.

Search by repository name

All content

Create a repository

Name	Last Pushed	Contains	Visibility	Scout
nimra07/program6	about 1 month ago	IMAGE	Public	Inactive
nimra07/prog6	about 1 month ago		Private	Inactive

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